SHYAM KUMAR

AI Security & AI Governance | Cybersecurity | NIST AI RMF · OWASP LLM Top 10 · MITRE ATLAS
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SUMMARY

CSE undergraduate working on AI security and governance: mapping real AI incidents across OWASP, MITRE ATLAS and the NIST AI RMF, and writing risk assessments that end in a named control. Identified that six recent AI-agent data-loss incidents had no attacker, a category ATLAS has no vocabulary for. Also builds hardware and full-stack products for international competitions.

AI SECURITY & GOVERNANCE PROJECTS

ai-incident-atlas | AI incident analysis — *Sep 2026 – Present*

- Mapped 10 real AI incidents across OWASP LLM Top 10, MITRE ATLAS (v5.6.0) and NIST AI RMF, each with the one control that would have stopped it
- Showed 6 agent data-loss incidents (Replit, Gemini CLI, Antigravity, Claude Code, Cursor, OpenClaw) had no adversary; sorted them into 5 failure mechanisms and proposed 5 testable checklist entries to run alongside ATLAS

NIST AI RMF assessment: AI coding agents with production access | ai-governance-portfolio — *Oct 2026*

- Rated the pattern High; 7-risk register mapped to RMF subcategories, 8 owned recommendations, and India DPDP Rules 2025 / CERT-In analysis showing accidental deletion is still a notifiable breach

redteam-log | AI red-teaming lab notebook — *Aug 2026 – Present*

- Mapped all 10 OWASP LLM Top 10 (2025) classes to ATLAS techniques with garak probes, controls and an RMF hook for each; garak + PyRIT + Ollama lab with a documented runbook

Kaaval | Permission layer for AI coding agents (Python) — *Oct 2026*

- Decides allow / ask / deny before an agent's shell, SQL or tool call runs; replays all 6 no-adversary incidents and refuses each while still allowing the narrow intended command; tamper-evident decision log; 39 tests, zero dependencies; runs live in the browser via Pyodide

Kaappu | Java TOTP authenticator, built from RFC 4226 / 6238 — *Oct 2026*

- Zero-dependency 2FA app: AES-256-GCM vault (PBKDF2, 600k iterations), Shamir secret-sharing backup of the vault key, replay-safe verifier; 70 tests incl. every RFC vector; written threat model (SECURITY.md)

FairDoor | Deadline-fair rate limiting, simulation — *Oct 2026*

- Per-student fair queuing with signed arrival stamps; in a 300-student shared-IP simulation, per-IP limiting left ~68 students past the deadline (worse than no limiter, ~60); FairDoor left 0

PRODUCT & HACKATHON PROJECTS

AquaNXT | ASU Venture X 2026 — *Mar 2026 – Present*

- Gravity water filters in three form factors (home, carry kit, school station) sharing one on-device monitoring assistant that reports only what it measures. Team selected as REC's finalist in the local round

UpAid | Cintana AI Challenge 2026 (team) — *Aug – Sep 2026*

- Evidence-first financial-aid finder for ASU-alliance campuses: every match cites the source document; registry of 34 campuses and 69 schemes, with human verification required before a policy is published

Virasat | PEC Hacks 4.0, Panimalar Engineering College (36-hour national hackathon) — *Aug 2026*

- 1st place, FinTech domain. Platform to find and claim unclaimed financial assets across RBI, EPFO, IRDAI, SEBI and LIC; rule-based document verification (file-type gating, duplicate detection, cross-check against asset records). TypeScript, Drizzle ORM, Vercel

LEADERSHIP & EXPERIENCE

Head, Tech & Skills Club | ASU Pathways (REC, Shiv Nadar University, Anurag University) — *Sep 2026 – Present*

- Founding head of a cross-university student club; built the club website, tech-skills-council.github.io

General Secretary | Velammal Bodhi Campus (Student Council) — *2023 – 2025*

- Hosted school-wide events and secured sponsorships as part of the elected student leadership team

COMPETITIONS

ASU Venture X 2026 (REC finalist, local round) · PEC Hacks 4.0 (1st, FinTech domain, Aug 2026) · Cintana AI Challenge 2026 · Pitchathon, Easwari Engineering College

EDUCATION

B.E. Computer Science and Engineering | Rajalakshmi Engineering College, ASU Pathways cohort — *Aug 2025 – May 2029* | SGPA: 7.5

High School, Mathematics and Biology (PCMB) | Velammal Bodhi Campus — *2023 – 2025* | 75% (First Class)

SKILLS

AI security / governance: OWASP LLM Top 10, MITRE ATLAS, NIST AI RMF, DPDP Act & Rules, threat modelling, prompt injection, garak, PyRIT
Programming: Java, Python, C, TypeScript, SQL, ESP32 / C++, Git
Languages: English (Cambridge C1 Advanced, score 187, 2024) · Tamil (native) · Hindi (DBHPS Prathamic) · French (studied through Class 10 board exams)

CERTIFICATIONS & COURSES

- Arizona State University: Co-Lecturing – Introduction to Engineering (Mar 2026)
- Anthropic (10+, Sep 2026): AI Fluency, Building with the Claude API, Model Context Protocol, Agents and Workflows, Prompt Engineering
- Microsoft AI Red Teaming 101 (completed, Oct 2026) · NIST AI Risk Management Framework (in progress)